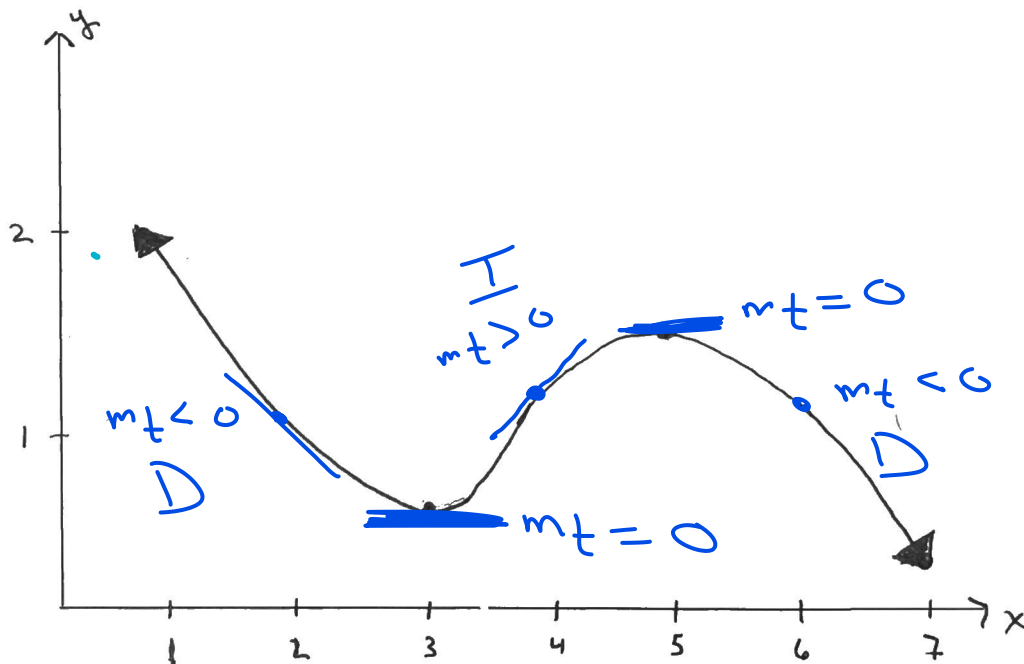


Unit 3 – Curve Sketching

Increasing & Decreasing Functions

- A function is **increasing** on an interval if the graph of the function **increases** from **left to right**.
- A function is **decreasing** on an interval if the graph of the function **decreases** from **left to right**.

Example: Locate the intervals of increase and decrease for the function below.



Increasing: $(3, 5)$

Decreasing: $(-\infty, 3) \cup (5, \infty)$

- A function is **increasing** on an interval if the function has a **positive** slope of the tangent on that interval.
- A function is **decreasing** on an interval if the function has a **negative** slope of the tangent on that interval.

Test for Increasing and Decreasing Functions

- If $f'(x) > 0$ for all x in an interval, then $f(x)$ is **increasing** in that interval.
- If $f'(x) < 0$ for all x in an interval, then $f(x)$ is **decreasing** in that interval.

Critical Numbers

- A **critical number**, c , is a number in the domain of $f(x)$ such that

$$f(x) = (x-1)^3$$



$$\underline{f'(c) = 0}$$

OR

$$\underline{f'(c) \text{ DNE.}}$$

$$\begin{aligned} -\frac{1}{x^2} &= 0 \\ -1 &= 0 \times \end{aligned}$$

critical #: NONE

$$f(x) = \frac{1}{x}$$

$$f'(x) = -\frac{1}{x^2}$$

$$\begin{aligned} &\sqrt{A} \\ &f'(0) \\ &\underline{\text{undefined}} \end{aligned}$$

Example: Determine the intervals on which the following functions are increasing or decreasing.

a) $f(x) = 3x^2 + 6x - 1$

$$f'(x) = 6x + 6$$

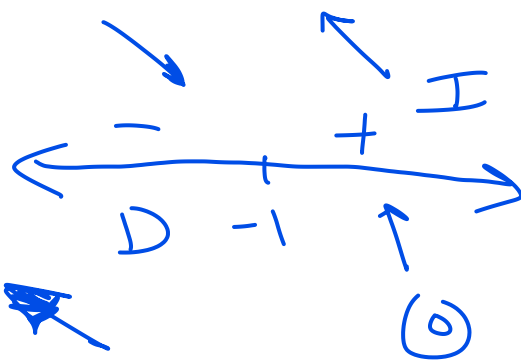
$$\therefore \text{critical \#s} \Rightarrow f'(x) = 0$$

$$6(x+1) = 0$$

$$x = -1 \quad (\text{critical \#})$$

since there is 1 critical # $\Rightarrow$ 2 intervals.

Interval	$(-\infty, -1)$	$(-1, \infty)$
$x+1$	-	+
$f'(x)$	-	+
$f(x)$		
	decreasing D	increasing I



$\therefore$ Decreasing $(-\infty, -1)$

Increasing $(-1, \infty)$

-1 not included.

b) $f(x) = x^3 + 9x^2 + 24x - 3$

$f'(x) = 3x^2 + 18x + 24$

$= 3(x^2 + 6x + 8)$

$\therefore$ critical #'s $\Rightarrow f'(x) = 0$

$3(x^2 + 6x + 8) = 0$

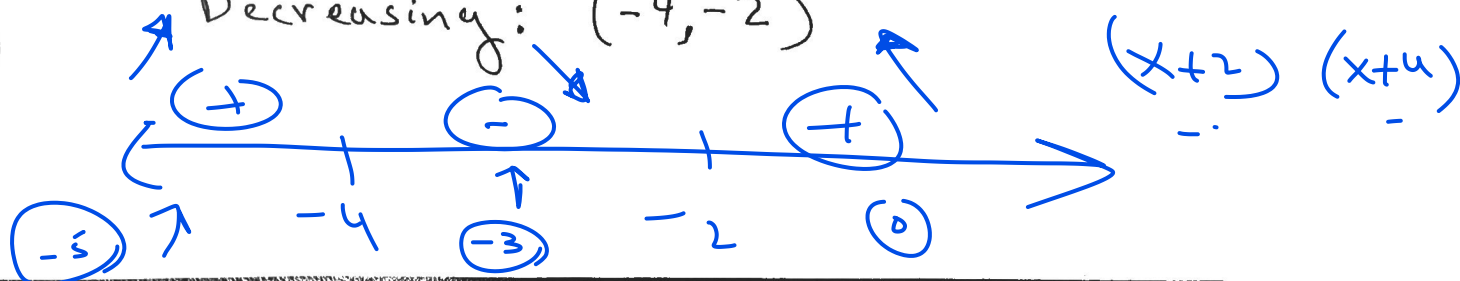
$3(x+4)(x+2) = 0$

$\therefore x = -4, -2 \Rightarrow 3$ intervals

Interval	$(-\infty, -4)$	$(-4, -2)$	$(-2, \infty)$
$x+4$	-	+	+
	$\times$	$\times$	$\times$
$x+2$	-	-	+
$f'(x)$	$\oplus$	$\ominus$	$\oplus$
$f(x)$	$\uparrow$	$\downarrow$	$\uparrow$
	increasing I	decreasing D	increasing I.

$\therefore$ Increasing: $(-\infty, -4) \cup (-2, \infty)$

Decreasing: $(-4, -2)$



$$c) f(x) = \frac{1}{4}x^4 + \frac{4}{3}x^3 - \frac{5}{2}x^2 \quad \leftarrow$$

$$f'(x) = \underline{x^3} + \underline{4x^2} - \underline{5x}$$

$$= x(\underline{x^2 + 4x - 5})$$

$$= x(\underline{x+5})(\underline{x-1})$$

critical #'s $\Rightarrow f'(x) = 0$

$\therefore x = 0, -5, 1 \Rightarrow 4 \text{ intervals}$

Interval	$(-\infty, -5)$	$(-5, 0)$	$(0, 1)$	$(1, \infty)$
x	-	-	+	+
$x+5$	$\times$	$\times$	$\times$	$\times$
$x-1$	$\times$	$\times$	$\times$	$\times$
$f'(x)$	$\ominus$	$\oplus$	$\ominus$	$\oplus$
$f(x)$	$\downarrow$ D	$\uparrow$ I	$\downarrow$ D	$\uparrow$ I

Increasing $(-5, 0) \cup (1, \infty)$

Decreasing $(-\infty, -5) \cup (0, 1)$

